

Preventing inlet unstart in air-breathing hypersonic vehicles using adaptive backstepping control with state constraints

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ARTICLE INFO

Keywords:

Air-breathing hypersonic vehicle
Inlet unstart
Adaptive dynamic inverse control
Adaptive backstepping control
Asymmetric barrier Lyapunov function
Symmetric barrier Lyapunov function

ABSTRACT

Aiming at the problem that the inlet is prone to unstart during the flight of air-breathing hypersonic vehicles (AHV), an adaptive backstepping control method with state constraints is proposed to prevent the occurrence of inlet unstart. Firstly, a parametric model of AHV with velocity, flight path angle, angle of attack and angle of attack rate as state variables is derived. Then, adaptive dynamic inverse control is adopted for the velocity subsystem, and adaptive backstepping control with state C is adopted for the flight path angle subsystem. The tracking errors of angle of attack and angle of attack rate are constrained by asymmetric barrier Lyapunov function (ABLF) and symmetric barrier Lyapunov function (SBLF) respectively, and smooth saturation function is constructed to restrict the virtual control commands. Adaptive laws are used to handle parameter uncertainty. Finally, it is proved by Lyapunov theory that the states of the closed-loop system can be uniformly and ultimately bounded, and the constraints on angle of attack and angle of attack rate can be satisfied. The simulation results show that compared with the adaptive backstepping control method, the proposed method can guarantee that the angle of attack and angle of attack rate are always within the operating limits of scramjet engine, and effectively avoid the occurrence of inlet unstart.

1. Introduction

The air-breathing hypersonic vehicle (AHV) has great military value and potential economic value because of its unique advantages of fast flight speed, long range, strong penetration ability and low cost-effectiveness. The normal operation of the scramjet engine is a significant guarantee for the AHV to accomplish the flight mission. As an important component of the scramjet, the working status of the hypersonic inlet greatly affects the success or failure of the whole mission.

There are two status of hypersonic inlet: inlet start and inlet unstart. Inlet start means that the internal flow field in the inlet has no effect on the flow capture characteristics of the inlet. Conversely is inlet unstart. The main feature of hypersonic inlet unstart is that the terminal normal shock waves in the inlet are pushed out of the inlet, and a strong shock wave system appears near the entrance of the inlet, which makes it impossible to organize normal supersonic flow in the inlet. At this time, the flow capture and total pressure recovery of the inlet decrease sharply, the flow quality deteriorates, and the engine experiences an instantaneous loss of thrust. Factors such as fuel supply of scramjet engine, back pressure of inlet exit, Mach number, angle of attack and angle

of attack rate affect the inlet start performance. The following mainly discuss the effect of angle of attack and angle of attack rate.

With the increase of the angle of attack, the total pressure recovery and kinetic efficiency of the inlet gradually decrease, and the flow capture gradually increases. When the angle of attack increases to a certain value, oblique shocks are formed at the entrance of the inlet, the flow separation appears behind the shock waves, the performance parameters change suddenly, and the inlet unstart occurs. The angle of attack at this time is called the unstart angle of attack, which is the upper bound of the normal operation of the scramjet engine. With the decrease of the angle of attack, the air inflow of the inlet decreases. So lower bound on the angle of attack arise from the necessity of maintaining a minimal air inflow due to normal combustion considerations. Therefore, the constraints of the normal operation of the scramjet on the angle of attack is asymmetric.

When the angle of attack changes dynamically, due to the aerodynamic hysteresis effect caused by the compressibility and viscosity of the airflow, there is a certain hysteresis phenomenon in the flow field characteristics and performance parameters of the hypersonic inlet, and the hysteresis characters between different performance parameters are

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<https://doi.org/10.1016/j.actaastro.2023.06.043>

Received 9 May 2022; Received in revised form 18 May 2023; Accepted 25 June 2023

Available online 3 July 2023

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different. For the problem of inlet unstart caused by the variation of angle of attack, with the increase of angle of attack rate, the unstart angle of attack value becomes larger, and the restart angle of attack value decreases, that is, the dynamic variation of angle of attack has a certain hysteresis effect on the process of inlet start [1].

Therefore, the control system of scramjet-powered air-breathing hypersonic vehicles should not only ensure attitude stability and trajectory tracking, but also constrain the angle of attack and angle of attack rate to permitted range.

Constraint control technique is an effective way to handle constraints. Constraint control methods mainly include reference generator, bounding functions and barrier Lyapunov function. Serrani et al. [2] incorporated a dynamic reference generator in the velocity loop of a nonlinear adaptive reconfigurable controller to ensure the fuel-to-air equivalence ratio (FER) command within the admissible bounds of operability of the scramjet engine. Famularo et al. [3] combined nonlinear dynamic inversion adaptive controller and state constraint mechanism based on bounding functions to limit the angle of attack and sideslip angle to permitted range of operating conditions of scramjet engine. Ngo et al. [4] firstly tackled the problem of multiple state constraints of a conventionally configured aircraft with barrier Lyapunov function in the backstepping paradigm. Tee et al. extended Ngo's barrier Lyapunov function to output-constrained nonlinear systems [5], time-varying output-constrained nonlinear systems [6], partial state-constrained nonlinear systems [7] and full state-constrained nonlinear systems [8]. Jin [9] incorporated a tan-type barrier Lyapunov function into adaptive backstepping control for a class of output-constrained multi-input single-output nonlinear systems.

The adaptive backstepping method combines the advantages of adaptive control and backstepping control, and has been widely used in the control of air-breathing hypersonic vehicles [10,11]. Barrier Lyapunov function is recently incorporated with adaptive backstepping method to solve the problem of uncertainty, strong coupling and state constraint of air-breathing hypersonic vehicles [12,13]. limited velocity, flight path angle, pitch angle and pitch angle rate based on the winged-cone model, without regard to the coupling of fuselage and scramjet engine [14–16]. indirectly implemented the constraints on the angle of attack by restricting flight path angle and pitch angle, while the equation relationship between these three angles $\alpha = \theta - \gamma$ was true when the aircraft only flew in the longitudinal motion, and this assumption was invalid in practical flight [17,18]. modeled the inlet unstart bounds of angle of attack as the function of velocity, and barrier Lyapunov function was used to limit angle of attack in the permissible set [19,20]. directly limited angle of attack and the constraints on angle of attack was symmetric, while the constraints on angle of attack required by scramjet engine was asymmetric in real flight [12–16]. constrained pitch angle rate, while the state that really affected the inlet performance was angle of attack rate rather than pitch angle rate. To the best of the author's knowledge, the above papers have not considered the constraints on the angle of attack rate.

Motivated by the previous discussions, this paper incorporates the limits of angle of attack and angle of attack rate in adaptive backstepping control for air-breathing hypersonic vehicle. For the first time, the angle of attack rate rather than pitch angle rate is used as the state variable in the parameterized model, preparing for imposing constraints on angle of attack rate in control algorithm. In order to ensure the normal operation of the scramjet engine, the barrier Lyapunov function and the smooth saturation function are combined to constrain the angle of attack and the angle of attack rate of the AHV. Compared with the existing literatures, the innovations of this paper are summarized as

follows.

- 1) A parameterized model with velocity, flight path angle, angle of attack and angle of attack rate as state variables is firstly derived for AHV. Previous papers have taken pitch angle rate as the state, while the state that really affects the inlet performance is angle of attack rate rather than pitch angle rate, so the parameterized model of AHV with angle of attack rate as state variable is derived in this paper.
- 2) The symmetric barrier Lyapunov function and the symmetric smooth saturation function are integrated with adaptive backstepping control to constrain angle of attack rate to admissible range. The angle of attack rate tracking error is constrained by the symmetric barrier Lyapunov function, and the virtual angle of attack rate command is limited by the symmetric smooth saturation function which overcome the non-smoothness of the saturation nonlinearity. Previous papers have not considered constraints on the angle of attack rate.
- 3) Different from the constraints imposed on the angle of attack rate, the constraints on the angle of attack are asymmetric, which is implemented by using asymmetric barrier Lyapunov function and asymmetric smooth saturation function. The angle of attack tracking error is constrained by the asymmetric barrier Lyapunov function, and the virtual angle of attack command is limited by the asymmetric smooth saturation function. In previous papers the constraints on the angle of attack are symmetric.

The paper is organized as follows: Section 2 describes the model of air-breathing hypersonic vehicles and establishes the parameterization model. Section 3 proposes an adaptive backstepping control method with state constraints. Section 4 presents simulation and comparison results. Finally, Section 5 gives conclusions.

2. Control objective and model formulation

2.1. The model of air-breathing hypersonic vehicles

In this paper, the control-oriented model (COM) of the quasi wave-rider hypersonic vehicle proposed by Parker [21] is used as the control object. The COM with velocity, altitude, flight path angle, pitch angle, and pitch angle rate as state variables is formulated by:

$$\begin{aligned}\dot{V} &= \frac{T \cos \alpha - D}{m} - g \sin \gamma \\ \dot{h} &= V \sin \gamma \\ \dot{\gamma} &= \frac{T \sin \alpha + L}{mV} - \frac{g \cos \gamma}{V} \\ \dot{\theta} &= Q \\ \dot{Q} &= \frac{M}{I_{yy}}\end{aligned}\quad (1)$$

Replacing pitch angle and pitch angle rate with angle of attack and angle of attack rate, the COM is rewritten as

$$\begin{aligned}\dot{V} &= \frac{T \cos \alpha - D}{m} - g \sin \gamma \\ \dot{h} &= V \sin \gamma \\ \dot{\gamma} &= \frac{T \sin \alpha + L}{mV} - \frac{g \cos \gamma}{V} \\ \dot{\alpha} &= P \\ \dot{P} &= \frac{M}{I_{yy}} - \ddot{\gamma}\end{aligned}\quad (2)$$

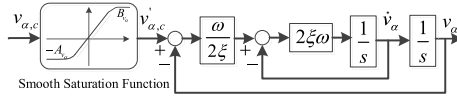


Fig. 1. Realization of asymmetric smooth saturation function and command filter.

where $V, h, \gamma, \vartheta, Q, \alpha$ and P represent velocity, altitude, flight path angle, pitch angle, pitch angle rate, angle of attack and angle of attack rate, respectively. I_{yy} and m are moments of inertia and mass of the hypersonic vehicle respectively. g is the acceleration of gravity. M, T, D , and L denote pitching moment, thrust, drag, and lift, respectively. The approximations of forces and moment are expressed as follows:

$$\begin{aligned} T &= \bar{q}S(C_{T,ER}(\alpha)ER + C_T(\alpha)) \\ D &= \bar{q}SC_D(\alpha) \\ L &= \bar{q}SC_L(\alpha) \\ M &= z_T T + \bar{q}SC_M(\alpha, \delta_e) \end{aligned} \quad (3)$$

where C_M, C_T, C_D , and C_L represent the coefficient of pitching moment, thrust, drag, and lift respectively. $S, \bar{c}, z_T$, and $\bar{q}$ denote reference area, reference length, thrust to moment coupling coefficient and dynamic pressure. ER is equivalency ratio, and δ_e is elevator deflection angle.

$$\bar{q} = \frac{1}{2} \rho V^2 \quad (4)$$

$$\rho = \rho_0 \exp[-(h - h_0)/h_s] \quad (5)$$

where ρ denotes air density.

Without considering the influence of elevator deflection angle on the lift coefficient and drag coefficient, the coefficients of longitudinal forces and moment are as follows:

$$\begin{aligned} C_{T,ER}(\alpha) &= C_{T,ER}^0 + C_{T,ER}^1 \alpha + C_{T,ER}^2 \alpha^2 + C_{T,ER}^3 \alpha^3 \\ C_T(\alpha) &= C_T^0 + C_T^1 \alpha + C_T^2 \alpha^2 + C_T^3 \alpha^3 \\ C_D(\alpha) &= C_D^0 + C_D^1 \alpha + C_D^2 \alpha^2 \\ C_L(\alpha) &= C_L^0 + C_L^1 \alpha \\ C_M(\alpha, \delta_e) &= C_M^0 + C_M^1 \alpha + C_M^2 \alpha^2 + C_M^3 \delta_e \end{aligned} \quad (6)$$

$$\begin{aligned} \ddot{\gamma} &= (\dot{\gamma})' = (f_{\gamma} + \Theta_{\gamma,1}^T \Phi_{\gamma,1} + \Theta_{\gamma,2}^T \Phi_{\gamma,2} \alpha)' = \left(-\frac{g \cos \gamma}{V} \right)' + \frac{S}{m} \left(\left(\frac{\bar{q}}{V} \right)' C_L^0 + \left(\frac{\bar{q}}{V} \right)' (C_{T,ER}(\alpha_{trim})ER + C_T(\alpha_{trim}) + C_L^1 \alpha) \right. \\ &\quad \left. + \frac{\bar{q}}{V} (C_{T,ER}(\alpha_{trim})(ER\alpha + ER\dot{\alpha}) + (C_T(\alpha_{trim}) + C_L^1 \alpha)\dot{\alpha}) \right) = \frac{g(V\dot{\gamma} \sin \gamma + \dot{V} \cos \gamma)}{V^2} + \frac{S}{m} (C_{T,ER}(\alpha_{trim}), C_{T,ER}(\alpha_{trim}), C_T(\alpha_{trim}) + C_L^1 \alpha) \\ &\quad \left(\begin{array}{c} ER\alpha \\ ER\dot{\alpha} \\ \dot{\alpha} \end{array} \right) \frac{\bar{q}}{V} + \frac{S}{m} (C_{T,ER}(\alpha_{trim}), C_T(\alpha_{trim}) + C_L^1 \alpha, C_L^0) \left(\begin{array}{c} ER\alpha \\ \alpha \\ 1 \end{array} \right) \left(\frac{\bar{q}}{V} \right)' \end{aligned} \quad (10)$$

2.2. The establishment of parameterization model

Substituting (3) (4) (5) (6) into (2) and referring to the parameterization process of flight path angle dynamics (the use of trim angle of attack) by An et al. [22], the parameterized model of the first, third and fourth equation of (2) is:

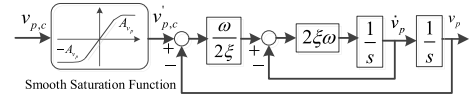


Fig. 2. Realization of symmetric smooth saturation function and command filter.

$$\begin{aligned} \dot{V} &= f_V + \Theta_{V,1}^T \Phi_{V,1} + \Theta_{V,2}^T \Phi_{V,2} ER \\ \dot{\gamma} &= f_{\gamma} + \Theta_{\gamma,1}^T \Phi_{\gamma,1} + \Theta_{\gamma,2}^T \Phi_{\gamma,2} \alpha \\ \dot{\alpha} &= P \end{aligned} \quad (7)$$

where

$$\begin{cases} f_V = -g \sin \gamma \\ \Theta_{V,1}^T = \frac{S}{m} (C_T^3, C_T^2, C_T^1, C_T^0, C_D^{\alpha^2}, C_D^{\alpha}, C_D^0) \\ \Phi_{V,1} = \bar{q}(\alpha^3 \cos \alpha, \alpha^2 \cos \alpha, \alpha \cos \alpha, \cos \alpha, -\alpha^2, -\alpha, -1)^T \\ \Theta_{V,2}^T = \frac{S}{m} (C_{T,ER}^3, C_{T,ER}^2, C_{T,ER}^1, C_{T,ER}^0) \\ \Phi_{V,2} = \bar{q}(\alpha^3, \alpha^2, \alpha, 1)^T \cos \alpha \end{cases} \quad (8)$$

$$\begin{cases} f_{\gamma} = -g \cos \gamma / V \\ \Theta_{\gamma,1}^T = \frac{S}{m} C_L^0 \\ \Phi_{\gamma,1} = \bar{q} / V \\ \Theta_{\gamma,2}^T = \frac{S}{m} (C_{T,ER}(\alpha_{trim}), C_L^{\alpha} + C_T(\alpha_{trim})) \\ \Phi_{\gamma,2} = \bar{q} / V (ER, 1)^T \end{cases} \quad (9)$$

with α_{trim} denotes trim angle of attack.

The parameterization process of $\ddot{\gamma}$ in the fifth equation of (2) is as follows:

Then, the parameterized model of the fifth equation of (2) is:

$$\dot{P} = f_P + \Theta_{P,1}^T \Phi_{P,1} + \Theta_{P,2}^T \Phi_{P,2} + \Theta_{P,3}^T \Phi_{P,3} + \Theta_{P,4}^T \Phi_{P,4} \delta_e \quad (11)$$

where

$$\begin{cases}
f_p = -g(V\dot{\gamma}\sin\gamma + \dot{V}\cos\gamma)/V^2 \\
\Theta_{P,1}^T = \frac{S}{I_{yy}} \left(C_{T,ER}^3, C_{T,ER}^2, C_{T,ER}^1, C_{T,ER}^0, C_{T,Z_T}^3, C_{T,Z_T}^2, C_{T,Z_T}^1 + C_M^{\alpha^2}\bar{c}, C_{T,Z_T}^0 + C_M^{\alpha}\bar{c}, C_{T,Z_T}^0 + C_M^0\bar{c} \right) \\
\Phi_{P,1} = \bar{q}(\alpha^3 ER, \alpha^2 ER, \alpha ER, ER, \alpha^3, \alpha^2, \alpha, 1)^T \\
\Theta_{P,2}^T = -\frac{S}{m} (C_{T,ER}(\alpha_{trim}), C_{T,ER}(\alpha_{trim}), C_T(\alpha_{trim}) + C_L^{\alpha}) \\
\Phi_{P,2} = \bar{q}/V(E\dot{R}\alpha, ER\dot{\alpha}, \dot{\alpha})^T \\
\Theta_{P,3}^T = -\frac{S}{m} (C_{T,ER}(\alpha_{trim}), C_{T,ER}(\alpha_{trim}) + C_L^{\alpha}, C_L^0) \\
\Phi_{P,3} = \left(\frac{\bar{q}}{V}\right)' (ER\alpha, \alpha, 1)^T \\
\Theta_{P,4}^T = \bar{c}SC_M^{\delta_e}/I_{yy} \\
\Phi_{P,4} = \bar{q}
\end{cases} \quad (12)$$

with $\left(\frac{\bar{q}}{V}\right)' = \frac{\rho}{2} \left(-\frac{\dot{h}V}{h} + \dot{V} \right)$.

Combining (7) and (11), the parametric model in strict feedback form of (2) is obtained as follows

$$\begin{aligned}
\dot{V} &= f_V + \Theta_{V,1}^T \Phi_{V,1} + \Theta_{V,2}^T \Phi_{V,2} ER \\
\dot{\gamma} &= f_{\gamma} + \Theta_{\gamma,1}^T \Phi_{\gamma,1} + \Theta_{\gamma,2}^T \Phi_{\gamma,2} \alpha \\
\dot{\alpha} &= P \\
\dot{P} &= f_P + \Theta_{P,1}^T \Phi_{P,1} + \Theta_{P,2}^T \Phi_{P,2} + \Theta_{P,3}^T \Phi_{P,3} + \Theta_{P,4}^T \Phi_{P,4} \delta_e
\end{aligned} \quad (13)$$

It can be decoupled into velocity subsystem and flight path angle subsystem.

Velocity subsystem:

$$\dot{V} = f_V + \Theta_{V,1}^T \Phi_{V,1} + \Theta_{V,2}^T \Phi_{V,2} ER \quad (14)$$

flight path angle subsystem:

$$\begin{aligned}
\dot{\gamma} &= f_{\gamma} + \Theta_{\gamma,1}^T \Phi_{\gamma,1} + \Theta_{\gamma,2}^T \Phi_{\gamma,2} \alpha \\
\dot{\alpha} &= P \\
\dot{P} &= f_P + \Theta_{P,1}^T \Phi_{P,1} + \Theta_{P,2}^T \Phi_{P,2} + \Theta_{P,3}^T \Phi_{P,3} + \Theta_{P,4}^T \Phi_{P,4} \delta_e
\end{aligned} \quad (15)$$

3. Controller design and analysis

The following reasonable assumptions are required for the design and analysis of the controller.

Assumption 1. The initial values of the angle of attack and angle of attack rate of the AHV are within the permitted range of the normal operation of the scramjet engine. That is

$$-A_{\alpha} \leq \alpha(0) \leq B_{\alpha}, |P(0)| \leq A_P.$$

where $-A_{\alpha}$ and B_{α} represent the lower and upper bounds of the angle of attack respectively, and A_P denotes the bound of the angle of attack rate.

Assumption 2. $\hat{\Theta}_{V,2}^T \Phi_{V,2}$, $\hat{\Theta}_{\gamma,2}^T \Phi_{\gamma,2}$ and $\hat{\Theta}_{P,4}^T \Phi_{P,4}$ are bounded and non-zero.

In order to facilitate the design and stability analysis of the controller, two useful lemmas are given without proof.

Lemma1. For any $x \in [1, \infty)$, the following inequality holds

$$\ln x \leq x - 1 \quad (16)$$

where the equal sign is established at $x = 1$.

Lemma2. For any $r \in R^m$, its estimated value $\hat{r}$ and the estimation error $\tilde{r} = r - \hat{r}$, the following inequality holds

$$\tilde{r}^T \hat{r} \leq \frac{1}{2} \|r\|^2 - \frac{1}{2} \|\tilde{r}\|^2 \quad (17)$$

3.1. Velocity tracking controller design

In this section, an adaptive dynamic inverse controller is designed for velocity subsystem to track velocity command V_c .

Velocity tracking error z_V is defined as

$$z_V = V - V_c \quad (18)$$

Then, velocity tracking error dynamics is

$$\dot{z}_V = f_V + \Theta_{V,1}^T \Phi_{V,1} + \Theta_{V,2}^T \Phi_{V,2} ER - \dot{V}_c \quad (19)$$

The control law for ER is designed as

$$ER = \frac{1}{\hat{\Theta}_{V,2}^T \Phi_{V,2}} \left(-k_V z_V - f_V + \dot{V}_c - \hat{\Theta}_{V,1}^T \Phi_{V,1} \right) \quad (20)$$

Where $k_V > 0$ is control parameter to be designed.

The aerodynamic coefficient and thrust coefficient C^* are calculated values or experimental values that differ from the true values, so adaptive laws is designed to estimate $\Theta_{*,*}$ online. The errors between real values $\Theta_{*,*}$ and estimated values $\hat{\Theta}_{*,*}$ are $\tilde{\Theta}_{*,*} = \Theta_{*,*} - \hat{\Theta}_{*,*}$.

$\hat{\Theta}_{V,1}$ and $\hat{\Theta}_{V,2}$ satisfy the following adaptive update law:

$$\begin{aligned}
\dot{\hat{\Theta}}_{V,1} &= H_{V,1}^{-1} (\Phi_{V,1} z_V - p_{V,1} \hat{\Theta}_{V,1}) \\
\dot{\hat{\Theta}}_{V,2} &= H_{V,2}^{-1} (\Phi_{V,2} z_V ER - p_{V,2} \hat{\Theta}_{V,2})
\end{aligned} \quad (21)$$

With $H_{V,1} > 0$, $H_{V,2} > 0$, $p_{V,1} > 0$, $p_{V,2} > 0$.

The positive definite function is selected as

$$L_V = \frac{1}{2} z_V^2 + \frac{1}{2} \|\tilde{\Theta}_{V,1}\|^2 H_{V,1} + \frac{1}{2} \|\tilde{\Theta}_{V,2}\|^2 H_{V,2} \quad (22)$$

The derivative of L_V is

$$\begin{aligned}
\dot{L}_V &= z_V \dot{z}_V - \tilde{\Theta}_{V,1}^T H_{V,1} \dot{\hat{\Theta}}_{V,1} - \tilde{\Theta}_{V,2}^T H_{V,2} \dot{\hat{\Theta}}_{V,2} \\
&= z_V \dot{f}_V + z_V \tilde{\Theta}_{V,1}^T \Phi_{V,1} + z_V \tilde{\Theta}_{V,2}^T \Phi_{V,2} ER - z_V \dot{V}_c + p_{V,1} \tilde{\Theta}_{V,1}^T \hat{\Theta}_{V,1} + p_{V,2} \tilde{\Theta}_{V,2}^T \hat{\Theta}_{V,2} \\
&= -k_V z_V^2 + p_{V,1} \tilde{\Theta}_{V,1}^T \hat{\Theta}_{V,1} + p_{V,2} \tilde{\Theta}_{V,2}^T \hat{\Theta}_{V,2}
\end{aligned} \quad (23)$$

3.2. Altitude tracking controller design

In this section, an adaptive backstepping controller with state constraints is designed for altitude subsystem to track altitude command h_c and ensure $-A_\alpha \leq \alpha \leq B_\alpha$, $|P| \leq A_P$ hold.

v_α and v_p denote virtual control commands for angle of attack and angle of attack rate respectively. γ_c is the flight path angle command. Tracking errors are defined as $z_h = h - h_c$, $z_\gamma = \gamma - \gamma_c$, $z_\alpha = \alpha - v_\alpha$, $z_p = P - v_p$.

Suppose the virtual controls satisfy

$$-A_{v_\alpha} \leq v_\alpha \leq B_{v_\alpha}, |v_p| \leq A_{v_p} \quad (24)$$

where $-A_{v_\alpha}$ and B_{v_α} represent the lower and upper bounds of v_α respectively, and A_{v_p} denotes the bound of v_p . If we can design a controller so that the tracking errors satisfy $-A_{z_\alpha} \leq z_\alpha \leq B_{z_\alpha}$, $|z_p| \leq A_{z_p}$ and inequalities $A_{v_\alpha} + A_{z_\alpha} \leq A_\alpha$, $B_{v_\alpha} + B_{z_\alpha} \leq B_\alpha$, $A_{v_p} + A_{z_p} \leq A_P$ hold, then the state constraints on angle of attack and angle of attack rate can be guaranteed. $-A_{z_\alpha}$ and B_{z_α} represent the lower and upper bounds of the tracking error of angle of attack z_α respectively, and A_{z_p} denotes the bound of tracking error of angle of attack rate z_p .

To prevent differential explosions caused by backstepping, a command filter is introduced in controller design. In order to ensure that (24) is true and that the differentials of the virtual commands are continuous, a smooth saturation function [23] is used to limit the virtual commands.

The nominal virtual command of angle of attack $v_{\alpha,c}$ is processed by the asymmetric smooth saturation function and command filter as shown in Fig. 1.

The asymmetric smooth saturation function in Fig. 1 is shown in

$$\begin{aligned} \dot{L}_\gamma &= z_\gamma^3 \dot{z}_\gamma - \tilde{\Theta}_{\gamma,1}^T H_{\gamma,1} \hat{\Theta}_{\gamma,1} - \tilde{\Theta}_{\gamma,2}^T H_{\gamma,2} \hat{\Theta}_{\gamma,2} = z_\gamma^3 f_\gamma + z_\gamma^3 \tilde{\Theta}_{\gamma,1}^T \Phi_{\gamma,1} + z_\gamma^3 \tilde{\Theta}_{\gamma,2}^T \Phi_{\gamma,2} \alpha - z_\gamma^3 \dot{\gamma}_c + p_{\gamma,1} \tilde{\Theta}_{\gamma,1}^T \hat{\Theta}_{\gamma,1} + p_{\gamma,2} \tilde{\Theta}_{\gamma,2}^T \hat{\Theta}_{\gamma,2} \\ &= z_\gamma^3 f_\gamma + z_\gamma^3 \tilde{\Theta}_{\gamma,1}^T \Phi_{\gamma,1} + z_\gamma^3 \left(-k_\gamma z_\gamma - f_\gamma + \dot{\gamma}_c - \tilde{\Theta}_{\gamma,1}^T \Phi_{\gamma,1} \right) + z_\gamma^3 \tilde{\Theta}_{\gamma,2}^T \Phi_{\gamma,2} \alpha - z_\gamma^3 \dot{\gamma}_c + p_{\gamma,1} \tilde{\Theta}_{\gamma,1}^T \hat{\Theta}_{\gamma,1} + p_{\gamma,2} \tilde{\Theta}_{\gamma,2}^T \hat{\Theta}_{\gamma,2} \\ &= -k_\gamma z_\gamma^4 + \tilde{\Theta}_{\gamma,2}^T \Phi_{\gamma,2} z_\gamma^3 z_\alpha + p_{\gamma,1} \tilde{\Theta}_{\gamma,1}^T \hat{\Theta}_{\gamma,1} + p_{\gamma,2} \tilde{\Theta}_{\gamma,2}^T \hat{\Theta}_{\gamma,2} \end{aligned} \quad (32)$$

(25):

$$v'_{\alpha,c} = \begin{cases} kB_{v_\alpha} + (B_{v_\alpha} - kB_{v_\alpha}) \tanh\left(\frac{v_{\alpha,c} - kB_{v_\alpha}}{B_{v_\alpha} - kB_{v_\alpha}}\right) & v_{\alpha,c} > kB_{v_\alpha} \\ v_{\alpha,c} - kA_{v_\alpha} \leq v_{\alpha,c} \leq kB_{v_\alpha} & \\ -kA_{v_\alpha} + (A_{v_\alpha} - kA_{v_\alpha}) \tanh\left(\frac{v_{\alpha,c} + kA_{v_\alpha}}{A_{v_\alpha} - kA_{v_\alpha}}\right) & v_{\alpha,c} < -kA_{v_\alpha} \end{cases} \quad (25)$$

The nominal virtual command of angle of attack rate $v_{p,c}$ is processed by the symmetric smooth saturation function and command filter as shown in Fig. 2.

The symmetric smooth saturation function in Fig. 2 is shown in (26).

$$v'_{p,c} = \begin{cases} kA_{v_p} + (A_{v_p} - kA_{v_p}) \tanh\left(\frac{v_{p,c} - kA_{v_p}}{A_{v_p} - kA_{v_p}}\right) & v_{p,c} > kA_{v_p} \\ v_{p,c} & |v_{p,c}| \leq kA_{v_p} \\ -kA_{v_p} + (A_{v_p} - kA_{v_p}) \tanh\left(\frac{v_{p,c} + kA_{v_p}}{A_{v_p} - kA_{v_p}}\right) & v_{p,c} < -kA_{v_p} \end{cases} \quad (26)$$

k in (25) and (26) in this paper is taken as 0.95.

γ_c can be described as

$$\gamma_c = \arcsin \frac{-k_h z_h - k_{hi} \int z_h + \dot{h}_c}{V} \quad (27)$$

Wherein: $k_h > 0$ and $k_{hi} > 0$ are the control parameters to be designed.

Each channel of the flight path angle subsystem is designed using the backstepping method, and the uncertain parameters are estimated using the adaptive update law. The design steps are as follows.

(1) Flight path angle tracking error dynamics is

$$\dot{z}_\gamma = f_\gamma + \Theta_{\gamma,1}^T \Phi_{\gamma,1} + \Theta_{\gamma,2}^T \Phi_{\gamma,2} \alpha - \dot{\gamma}_c = f_\gamma + \Theta_{\gamma,1}^T \Phi_{\gamma,1} + \Theta_{\gamma,2}^T \Phi_{\gamma,2} (v_\alpha + z_\alpha) - \dot{\gamma}_c \quad (28)$$

The control law for v_α is designed as

$$v_\alpha = \frac{1}{\tilde{\Theta}_{\gamma,2}^T \Phi_{\gamma,2}} \left(-k_\gamma z_\gamma - f_\gamma + \dot{\gamma}_c - \tilde{\Theta}_{\gamma,1}^T \Phi_{\gamma,1} \right) \quad (29)$$

Where $k_\gamma > 0$ is control parameter to be designed, $\hat{\Theta}_{\gamma,1}$, $\hat{\Theta}_{\gamma,2}$ satisfy the following adaptive update law:

$$\begin{aligned} \dot{\hat{\Theta}}_{\gamma,1} &= H_{\gamma,1}^{-1} \left(\Phi_{\gamma,1} z_\gamma^3 - p_{\gamma,1} \hat{\Theta}_{\gamma,1} \right) \\ \dot{\hat{\Theta}}_{\gamma,2} &= H_{\gamma,2}^{-1} \left(\Phi_{\gamma,2} z_\gamma^3 \alpha - p_{\gamma,2} \hat{\Theta}_{\gamma,2} \right) \end{aligned} \quad (30)$$

With $H_{\gamma,1} > 0$, $H_{\gamma,2} > 0$, $p_{\gamma,1} > 0$, $p_{\gamma,2} > 0$.

The positive definite function L_γ is selected as

$$L_\gamma = \frac{1}{4} z_\gamma^4 + \frac{1}{2} \|\tilde{\Theta}_{\gamma,1}\|^2 H_{\gamma,1} + \frac{1}{2} \|\tilde{\Theta}_{\gamma,2}\|^2 H_{\gamma,2} \quad (31)$$

The derivative of L_γ is

where the coupling term $\hat{\Theta}_{\gamma,2}^T \Phi_{\gamma,2} z_\gamma^3 z_\alpha$ is canceled in the subsequent step.

(2) Angle of attack tracking error dynamics is

$$\dot{z}_\alpha = P - \dot{v}_\alpha = v_p + z_p - \dot{v}_\alpha \quad (33)$$

An asymmetric barrier Lyapunov function (ABLF) [5] is defined as:

$$W_\alpha = \frac{q(z_\alpha)}{2} \ln \frac{B_{z_\alpha}^2}{B_{z_\alpha}^2 - z_\alpha^2} + \frac{1 - q(z_\alpha)}{2} \ln \frac{A_{z_\alpha}^2}{A_{z_\alpha}^2 - z_\alpha^2} \quad (34)$$

where $q(z_\alpha)$ is expressed as

$$q(z_\alpha) = \begin{cases} 1 & z_\alpha > 0 \\ 0 & z_\alpha \leq 0 \end{cases} \quad (35)$$

For convenience, we abbreviate $q(z_\alpha)$ by q , unless otherwise stated.

The derivative of W_α is

$$\dot{W}_\alpha = \tau_\alpha \dot{z}_\alpha \quad (36)$$

Where $\tau_\alpha = \left(\frac{q}{B_{z_\alpha}^2 - z_\alpha^2} + \frac{1-q}{A_{z_\alpha}^2 - z_\alpha^2} \right) z_\alpha$.

The control law for v_p is designed as

$$\dot{v}_p = -k_a z_a + \dot{v}_a - \hat{\Theta}_{\gamma,2}^T \Phi_{\gamma,2} z_\gamma^3 (q(B_{z_a}^2 - z_a^2) + (1-q)(A_{z_a}^2 - z_a^2)) \quad (37)$$

Where z_γ^3 instead of z_γ is to ensure that $\dot{v}_p$ is continuous. $k_a > 0$ is control parameter to be designed, and the last term on the right-hand side is to cancel the residual coupling term $\hat{\Theta}_{\gamma,2}^T \Phi_{\gamma,2} z_\gamma^3 z_a$ in (32).

The positive definite function L_a is selected as:

$$L_a = W_a = \frac{q}{2} \ln \frac{B_{z_a}^2}{B_{z_a}^2 - z_a^2} + \frac{1-q}{2} \ln \frac{A_{z_a}^2}{A_{z_a}^2 - z_a^2} \quad (38)$$

The derivative of L_a is

$$\begin{aligned} \dot{L}_a &= \left(\frac{q}{B_{z_a}^2 - z_a^2} + \frac{1-q}{A_{z_a}^2 - z_a^2} \right) z_a \dot{z}_a = \left(\frac{q}{B_{z_a}^2 - z_a^2} + \frac{1-q}{A_{z_a}^2 - z_a^2} \right) z_a \left(z_p - k_a z_a - \right. \\ &\quad \left. \hat{\Theta}_{\gamma,2}^T \Phi_{\gamma,2} z_\gamma^3 (q(B_{z_a}^2 - z_a^2) + (1-q)(A_{z_a}^2 - z_a^2)) \right) = \left(\frac{q}{B_{z_a}^2 - z_a^2} + \frac{1-q}{A_{z_a}^2 - z_a^2} \right) z_a z_p - \left(\frac{q}{B_{z_a}^2 - z_a^2} + \frac{1-q}{A_{z_a}^2 - z_a^2} \right) k_a z_a^2 - \hat{\Theta}_{\gamma,2}^T \Phi_{\gamma,2} z_\gamma^3 z_a \end{aligned} \quad (39)$$

where the coupling term $\left(\frac{q}{B_{z_a}^2 - z_a^2} + \frac{1-q}{A_{z_a}^2 - z_a^2} \right) z_a z_p$ is canceled in the subsequent step.

(3) Angle of attack rate tracking error dynamics is

$$\dot{z}_p = f_p + \hat{\Theta}_{p,1}^T \Phi_{p,1} + \hat{\Theta}_{p,2}^T \Phi_{p,2} + \hat{\Theta}_{p,3}^T \Phi_{p,3} + \hat{\Theta}_{p,4}^T \Phi_{p,4} \delta_e - \dot{v}_p \quad (40)$$

A symmetric barrier Lyapunov function (SBLF) [5] is defined as:

$$W_p = \frac{1}{2} \ln \frac{A_{z_p}^2}{A_{z_p}^2 - z_p^2} \quad (41)$$

The derivative of W_p is

$$\dot{W}_p = \tau_p \dot{z}_p \quad (42)$$

Where $\tau_p = z_p / (A_{z_p}^2 - z_p^2)$.

The control law for δ_e is designed as

$$\begin{aligned} \delta_e &= \frac{1}{\hat{\Theta}_{p,4}^T \Phi_{p,4}} \left(-k_p z_p - \left(\frac{q}{B_{z_a}^2 - z_a^2} + \frac{1-q}{A_{z_a}^2 - z_a^2} \right) (A_{z_p}^2 - z_p^2) z_a + \dot{v}_p \right. \\ &\quad \left. - f_p - \hat{\Theta}_{p,1}^T \Phi_{p,1} - \hat{\Theta}_{p,2}^T \Phi_{p,2} - \hat{\Theta}_{p,3}^T \Phi_{p,3} \right) \end{aligned} \quad (43)$$

Where $k_p > 0$ is control parameter to be designed, and the second term on the right-hand side is to cancel the residual coupling term $\left(\frac{q}{B_{z_a}^2 - z_a^2} + \frac{1-q}{A_{z_a}^2 - z_a^2} \right) z_a z_p$ in (39). $\hat{\Theta}_{p,1}$, $\hat{\Theta}_{p,2}$, $\hat{\Theta}_{p,3}$ and $\hat{\Theta}_{p,4}$ satisfy the following adaptive update law:

$$\begin{aligned} \dot{\hat{\Theta}}_{p,1} &= H_{p,1}^{-1} (\Phi_{p,1} \tau_p - p_{p,1} \hat{\Theta}_{p,1}) \\ \dot{\hat{\Theta}}_{p,2} &= H_{p,2}^{-1} (\Phi_{p,2} \tau_p - p_{p,2} \hat{\Theta}_{p,2}) \\ \dot{\hat{\Theta}}_{p,3} &= H_{p,3}^{-1} (\Phi_{p,3} \tau_p - p_{p,3} \hat{\Theta}_{p,3}) \\ \dot{\hat{\Theta}}_{p,4} &= H_{p,4}^{-1} (\Phi_{p,4} \tau_p \delta_e - p_{p,4} \hat{\Theta}_{p,4}) \end{aligned} \quad (44)$$

With $H_{p,1} > 0$, $H_{p,2} > 0$, $H_{p,3} > 0$, $H_{p,4} > 0$, $p_{p,1} > 0$, $p_{p,2} > 0$, $p_{p,3} > 0$, $p_{p,4} > 0$.

The positive definite function L_p is composed of symmetric barrier function and quadratic functions:

$$L_p = W_p + \frac{1}{2} \|\tilde{\Theta}_{p,1}\|^2 H_{p,1} + \frac{1}{2} \|\tilde{\Theta}_{p,2}\|^2 H_{p,2} + \frac{1}{2} \|\tilde{\Theta}_{p,3}\|^2 H_{p,3} + \frac{1}{2} \|\tilde{\Theta}_{p,4}\|^2 H_{p,4} \quad (45)$$

The derivative of L_p is

$$\begin{aligned} \dot{L}_p &= \tau_p \dot{z}_p - \tilde{\Theta}_{p,1}^T H_{p,1} \dot{\hat{\Theta}}_{p,1} - \tilde{\Theta}_{p,2}^T H_{p,2} \dot{\hat{\Theta}}_{p,2} - \tilde{\Theta}_{p,3}^T H_{p,3} \dot{\hat{\Theta}}_{p,3} - \tilde{\Theta}_{p,4}^T H_{p,4} \dot{\hat{\Theta}}_{p,4} \\ &= \tau_p f_p + \tau_p \hat{\Theta}_{p,1}^T \Phi_{p,1} + \tau_p \hat{\Theta}_{p,2}^T \Phi_{p,2} + \tau_p \hat{\Theta}_{p,3}^T \Phi_{p,3} + \tau_p \hat{\Theta}_{p,4}^T \Phi_{p,4} \delta_e - \tau_p \dot{v}_p \\ &\quad + p_{p,1} \tilde{\Theta}_{p,1}^T \hat{\Theta}_{p,1} + p_{p,2} \tilde{\Theta}_{p,2}^T \hat{\Theta}_{p,2} + p_{p,3} \tilde{\Theta}_{p,3}^T \hat{\Theta}_{p,3} + p_{p,4} \tilde{\Theta}_{p,4}^T \hat{\Theta}_{p,4} \\ &= \tau_p f_p + \tau_p \sum_{i=1}^3 \tilde{\Theta}_{p,i}^T \Phi_{p,i} + \tau_p \left(-k_p z_p - \left(\frac{q}{B_{z_a}^2 - z_a^2} + \frac{1-q}{A_{z_a}^2 - z_a^2} \right) (A_{z_p}^2 - z_p^2) z_a \right) \end{aligned}$$

$$\begin{aligned} &+ \dot{v}_p - f_p - \hat{\Theta}_{p,1}^T \Phi_{p,1} - \hat{\Theta}_{p,2}^T \Phi_{p,2} - \hat{\Theta}_{p,3}^T \Phi_{p,3} - \tau_p \dot{v}_p + \sum_{i=1}^4 p_{p,i} \tilde{\Theta}_{p,i}^T \hat{\Theta}_{p,i} \\ &= -\frac{k_p z_p}{A_{z_p}^2 - z_p^2} \left(\frac{q}{B_{z_a}^2 - z_a^2} + \frac{1-q}{A_{z_a}^2 - z_a^2} \right) z_a z_p + \sum_{i=1}^4 p_{p,i} \tilde{\Theta}_{p,i}^T \hat{\Theta}_{p,i} \end{aligned} \quad (46)$$

3.3. Performance analysis of the closed-loop system

Theorem1. Under Assumptions 1-2, consider the model consisting of (2) (3) (4) (5) (6), the control algorithms composed of the control laws ((20) (29) (37) (43)) and the update laws((21) (30) (44)) can ensure the following properties hold:

- (1) The angle of attack and angle of attack rate are always within the permitted range of scramjet engine;
- (2) All state tracking errors and parameter estimation errors ultimately converge into a bounded set.
- (3) All the closed-loop signals are uniformly bounded.

Proof The Lyapunov function candidates is selected as

$$L = L_v + L_\gamma + L_a + L_p \quad (47)$$

where L_v , L_γ , L_a and L_p are defined in (22), (31), (38) and (45), respectively. Differentiating L with respect to time:

Table 1
The parameters values of AHV in simulation.

Parameter	value	Parameter	value
C_L^0	-1.8714×10^{-2}	C_T^0	-2.7×10^{-3}
$C_L^0/1/\text{rad}$	4.6773	$C_T^0/1/\text{rad}$	-6.6×10^{-2}
C_M^0	1.8979×10^{-1}	$C_T^0/1/\text{rad}^2$	-4.712×10^{-1}
$C_M^0/1/\text{rad}$	2.1335	$C_T^0/1/\text{rad}^3$	-1.0152
$C_M^0/1/\text{rad}^2$	6.2926	$C_{T,ER}^0$	1.739×10^{-1}
$C_M^0/1/\text{rad}$	-1.2897	$C_{T,ER}^1/1/\text{rad}$	9.693×10^{-1}
C_D^0	1.0131×10^{-2}	$C_{T,ER}^2/1/\text{rad}^2$	7.312×10^{-1}
$C_D^0/1/\text{rad}$	-4.5315×10^{-2}	$C_{T,ER}^3/1/\text{rad}^3$	-1.02788×10^1
$C_D^0/1/\text{rad}^2$	5.8224		

Table 2
Initial conditions and state constraints.

Physical quantity	Symbol	Value
initial altitude/m	$h(0)$	24,613
initial velocity/m/s	$V(0)$	1981
initial flight path angle/deg	$\gamma(0)$	0
initial pitch angle/deg	$\theta(0)$	1.9
initial pitch rate/deg/s	$Q(0)$	3.9
lower bound of angle of attack/deg	$-A_\alpha$	-4
upper bound of angle of attack/deg	B_α	6
upper bound of absolute value of angle of attack rate/deg/s	A_P	5

Table 3
The parameters values of controller.

Parameter	value	Parameter	value
k_V	0.5	A_{V_α}/deg	1.5
k_h	0.15	B_{V_α}/deg	5.8
k_{hi}	0.015	$A_{V_P}/\text{deg/s}$	4.5
k_γ	0.5	A_{z_α}/deg	2.5
k_α	2	B_{z_α}/deg	0.2
k_P	10	$A_{z_P}/\text{deg/s}$	0.5

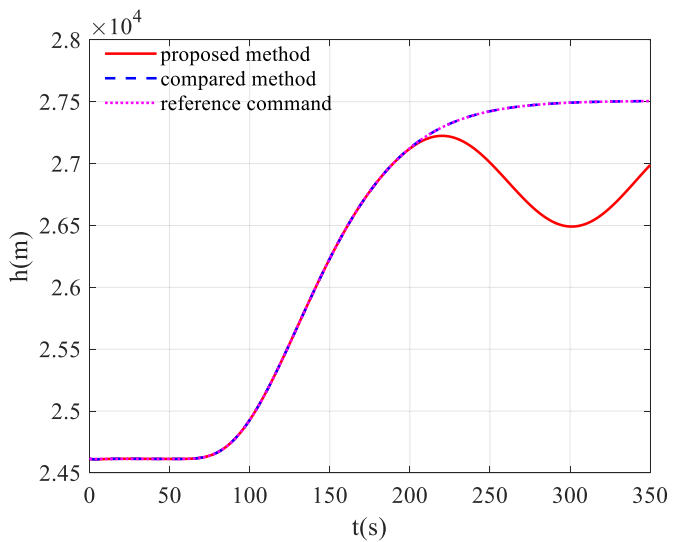


Fig. 3. Altitude tracking.

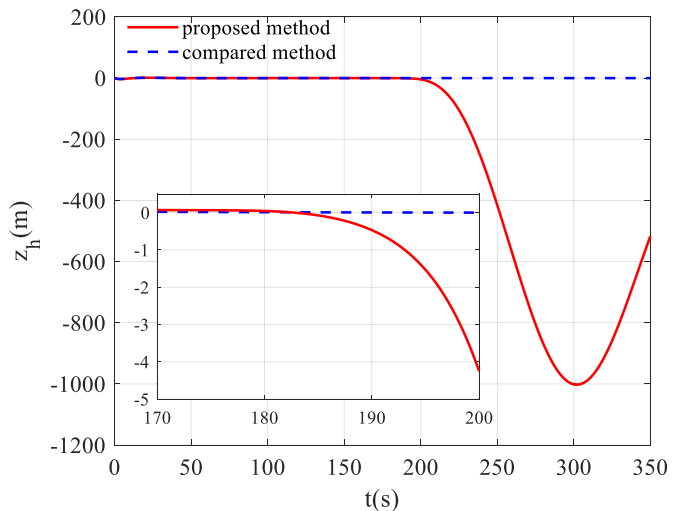


Fig. 4. Altitude tracking error.

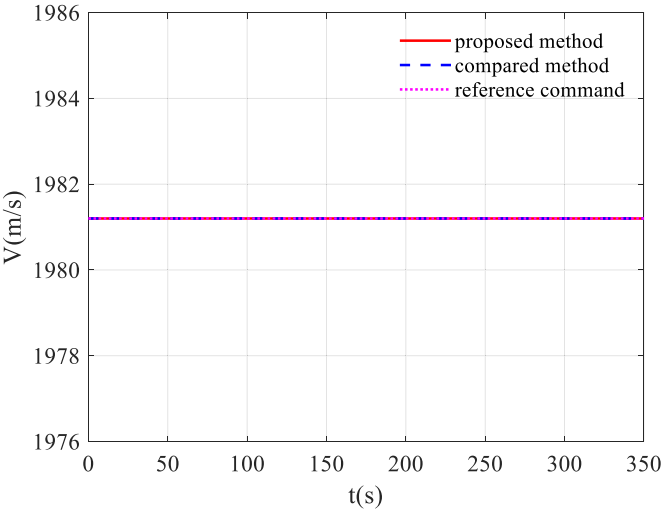


Fig. 5. Velocity tracking.

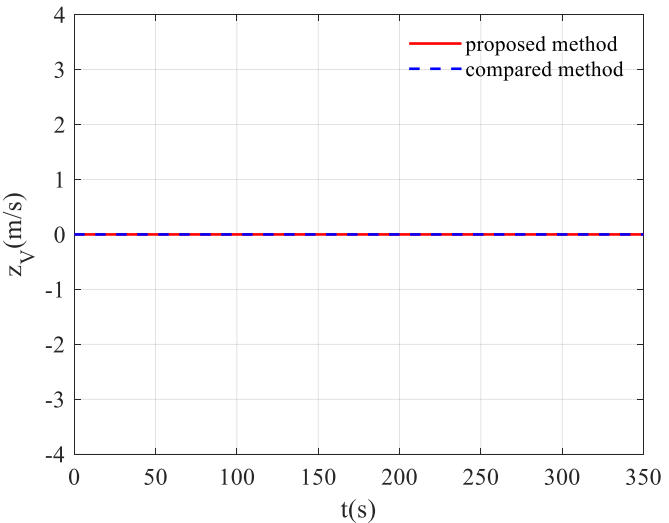


Fig. 6. Velocity tracking error.

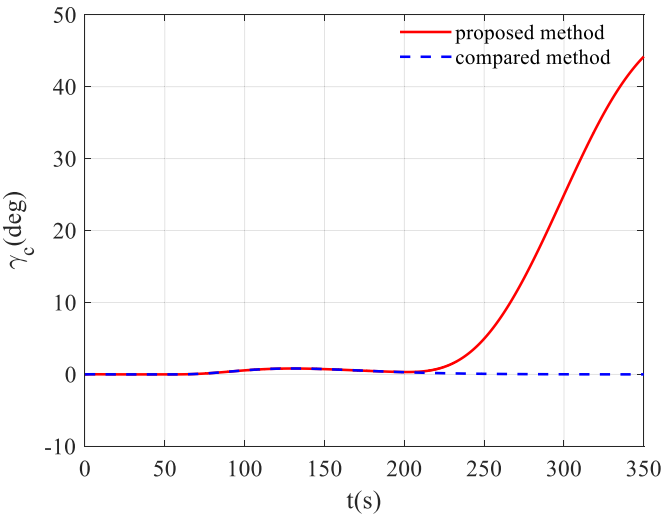


Fig. 7. Flight path angle command.

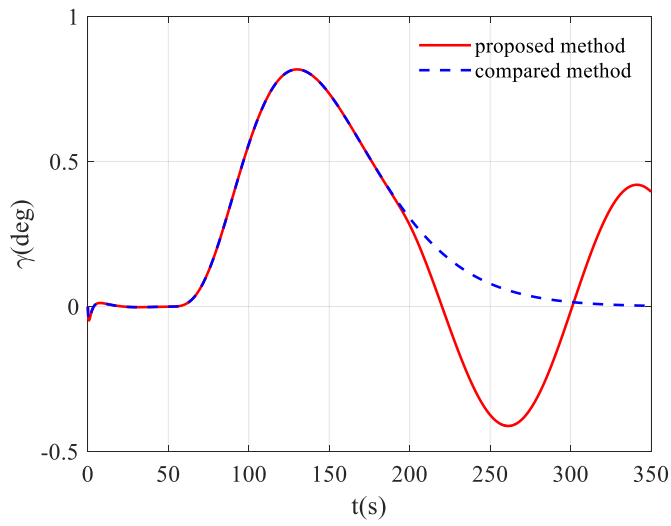


Fig. 8. Flight path angle.

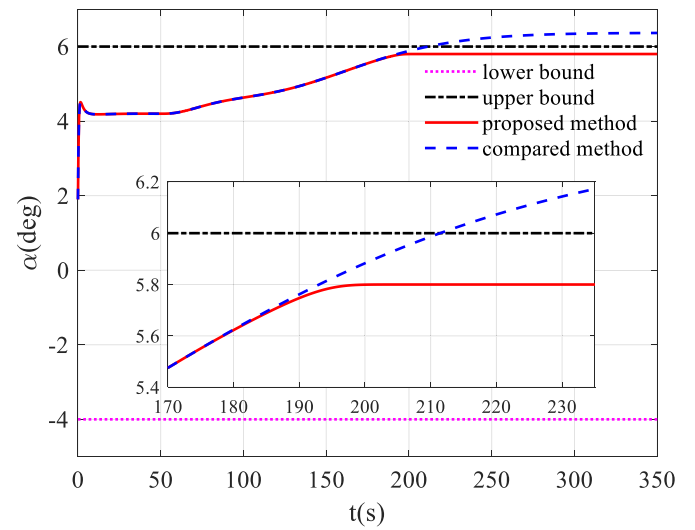


Fig. 11. Angle of attack.

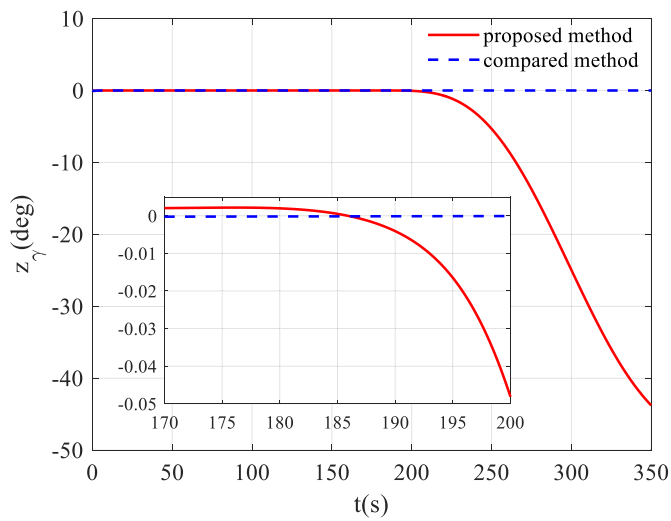


Fig. 9. Flight path angle tracking error.

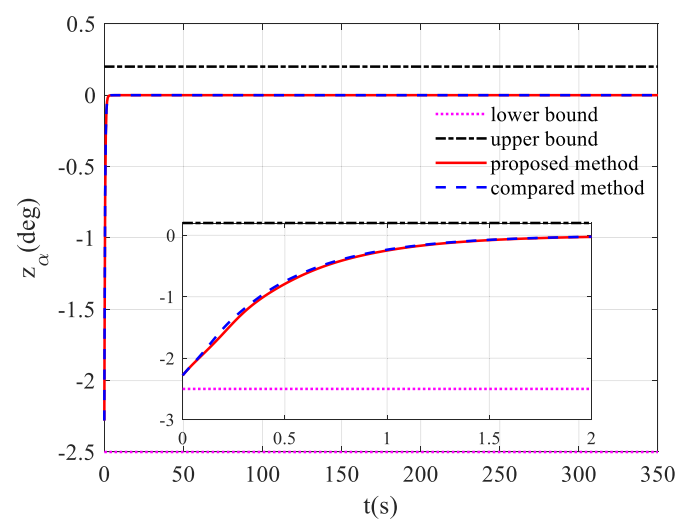


Fig. 12. Angle of attack tracking error.

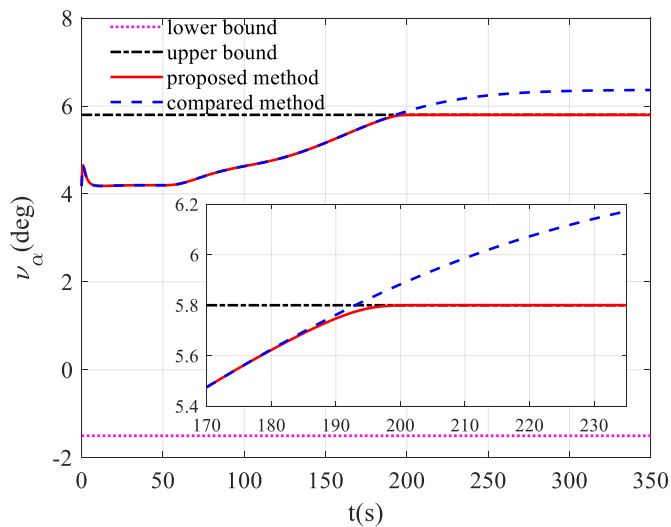


Fig. 10. Virtual angle of attack command.

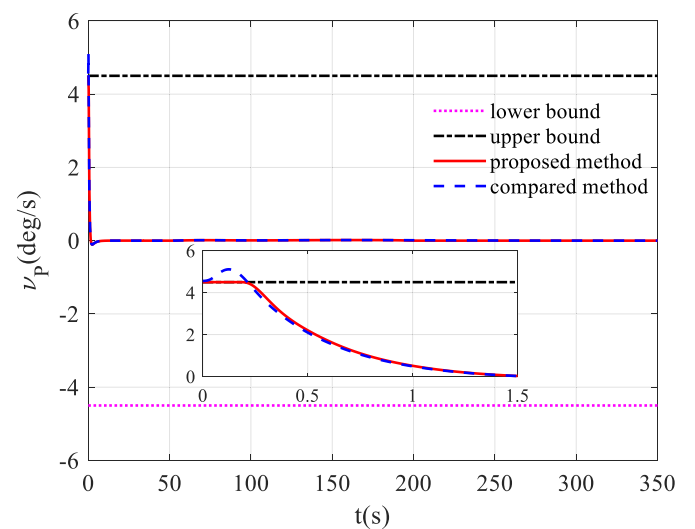


Fig. 13. Virtual angle of attack rate command.

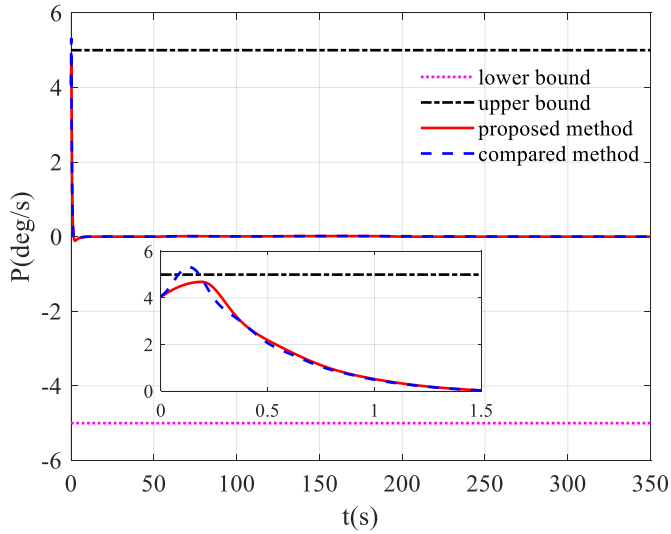


Fig. 14. Angle of attack rate.

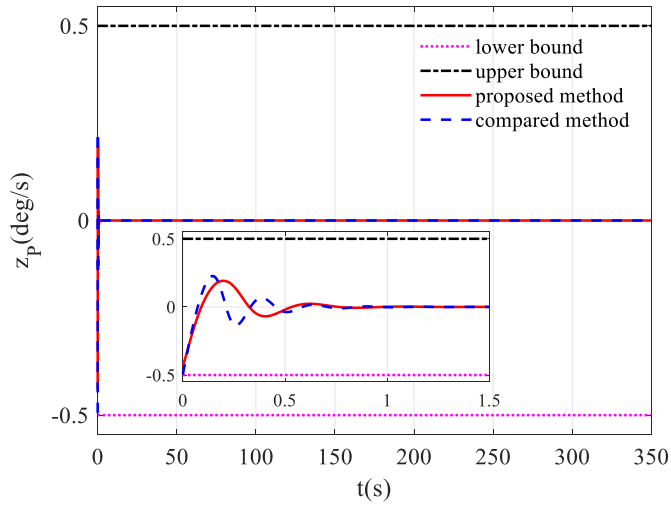


Fig. 15. Angle of attack rate tracking error.

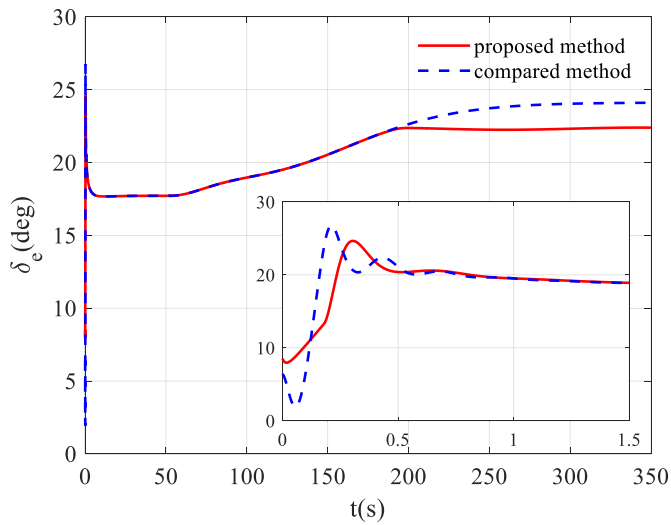


Fig. 16. Control command of elevator.

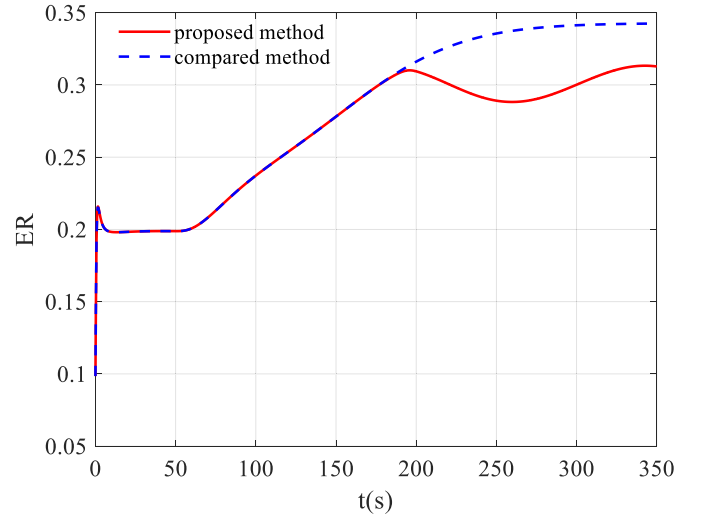
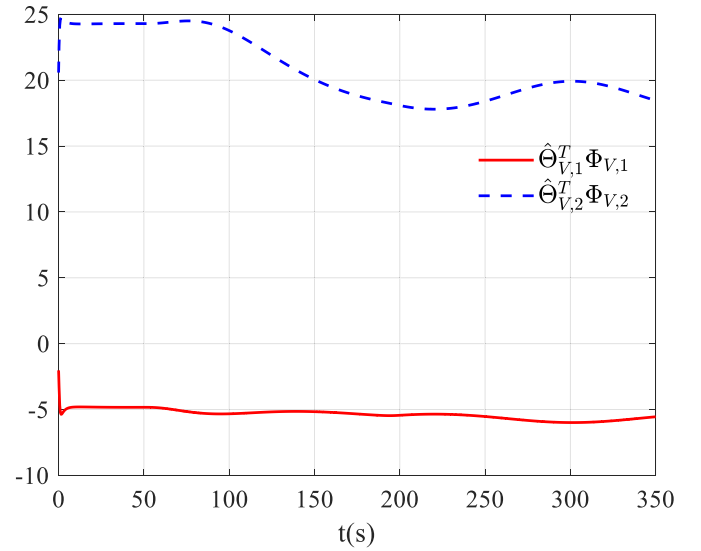


Fig. 17. Control command of equivalent ratio.

Fig. 18. $\hat{\Theta}_{V,1}^T \Phi_{V,1}$ and $\hat{\Theta}_{V,2}^T \Phi_{V,2}$.

$$\begin{aligned} \dot{L} = \dot{L}_V + \dot{L}_\gamma + \dot{L}_a + \dot{L}_p = & -k_V z_V^2 - k_\gamma z_\gamma^4 - \left(\frac{q}{B_{za}^2 - z_a^2} + \frac{1-q}{A_{za}^2 - z_a^2} \right) k_a z_a^2 - \frac{k_p z_p^2}{A_{zp}^2 - z_p^2} \\ & + \sum_{i=1}^2 p_{V,i} \tilde{\Theta}_{V,i}^T \hat{\Theta}_{V,i} + \sum_{i=1}^2 p_{\gamma,i} \tilde{\Theta}_{\gamma,i}^T \hat{\Theta}_{\gamma,i} + \sum_{i=1}^4 p_{p,i} \tilde{\Theta}_{p,i}^T \hat{\Theta}_{p,i} \end{aligned} \quad (48)$$

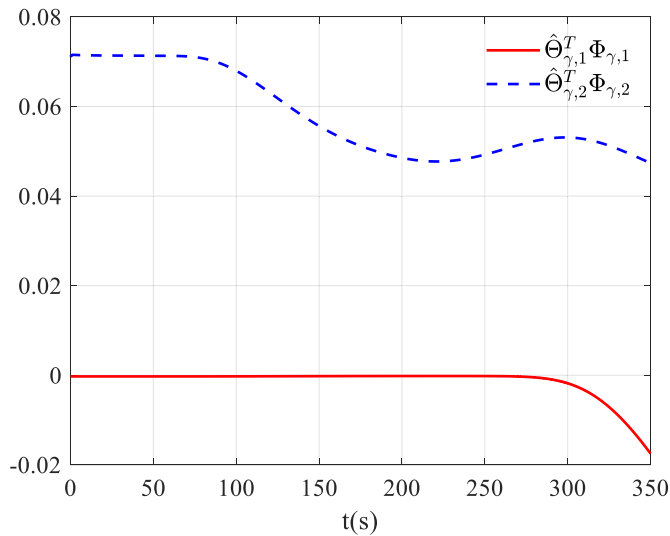
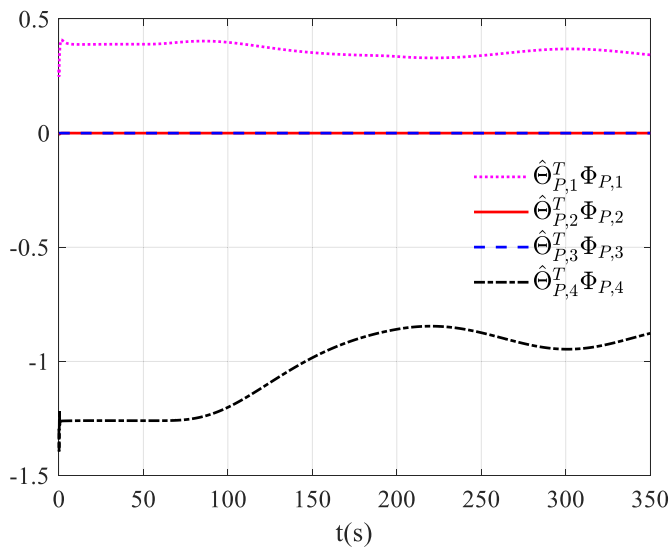
According to Lemma 2, the following inequalities holds:

$$\sum_{i=1}^2 p_{V,i} \tilde{\Theta}_{V,i}^T \hat{\Theta}_{V,i} \leq \sum_{i=1}^2 \left(\frac{p_{V,i}}{2} \Theta_{V,i}^2 - \frac{p_{V,i}}{2} \tilde{\Theta}_{V,i}^2 \right) \quad (49)$$

$$\sum_{i=1}^2 p_{\gamma,i} \tilde{\Theta}_{\gamma,i}^T \hat{\Theta}_{\gamma,i} \leq \sum_{i=1}^2 \left(\frac{p_{\gamma,i}}{2} \Theta_{\gamma,i}^2 - \frac{p_{\gamma,i}}{2} \tilde{\Theta}_{\gamma,i}^2 \right) \quad (50)$$

$$\sum_{i=1}^4 p_{p,i} \tilde{\Theta}_{p,i}^T \hat{\Theta}_{p,i} \leq \sum_{i=1}^4 \left(\frac{p_{p,i}}{2} \Theta_{p,i}^2 - \frac{p_{p,i}}{2} \tilde{\Theta}_{p,i}^2 \right) \quad (51)$$

Applying (49) (50) (51) to (48):

Fig. 19. $\hat{\Theta}_{\gamma,1}^T \Phi_{\gamma,1}$ and $\hat{\Theta}_{\gamma,2}^T \Phi_{\gamma,2}$.Fig. 20. $\hat{\Theta}_{P,1}^T \Phi_{P,1} \sim \hat{\Theta}_{P,4}^T \Phi_{P,4}$.

$$\begin{aligned} \dot{L} \leq & -k_V z_V^2 - k_\gamma z_\gamma^4 - \left(\frac{q}{B_{z_a}^2 - z_a^2} + \frac{1-q}{A_{z_a}^2 - z_a^2} \right) k_\alpha z_\alpha^2 - \frac{k_P z_P^2}{A_{z_P}^2 - z_P^2} \\ & - \sum_{i=1}^2 \frac{p_{V,i}}{2} \tilde{\Theta}_{V,i}^2 - \sum_{i=1}^2 \frac{p_{\gamma,i}}{2} \tilde{\Theta}_{\gamma,i}^2 - \sum_{i=1}^4 \frac{p_{P,i}}{2} \tilde{\Theta}_{P,i}^2 \\ & + \sum_{i=1}^2 \frac{p_{V,i}}{2} \Theta_{V,i}^2 + \sum_{i=1}^2 \frac{p_{\gamma,i}}{2} \Theta_{\gamma,i}^2 + \sum_{i=1}^4 \frac{p_{P,i}}{2} \Theta_{P,i}^2 \end{aligned} \quad (52)$$

Based on Lemma 1, we can further derive:

$$\begin{aligned} \dot{L} \leq & -k_V z_V^2 - k_\gamma z_\gamma^4 - k_\alpha \left(q \ln \frac{B_{z_a}^2}{B_{z_a}^2 - z_a^2} + (1-q) \ln \frac{A_{z_a}^2}{A_{z_a}^2 - z_a^2} \right) \\ & - k_P \ln \frac{A_{z_P}^2}{A_{z_P}^2 - z_P^2} - \sum_{i=1}^2 \frac{p_{V,i}}{2} \tilde{\Theta}_{V,i}^2 - \sum_{i=1}^2 \frac{p_{\gamma,i}}{2} \tilde{\Theta}_{\gamma,i}^2 - \sum_{i=1}^4 \frac{p_{P,i}}{2} \tilde{\Theta}_{P,i}^2 \\ & + \sum_{i=1}^2 \frac{p_{V,i}}{2} \Theta_{V,i}^2 + \sum_{i=1}^2 \frac{p_{\gamma,i}}{2} \Theta_{\gamma,i}^2 + \sum_{i=1}^4 \frac{p_{P,i}}{2} \Theta_{P,i}^2 \\ \leq & -\mu_1 L + \beta_1 \end{aligned} \quad (53)$$

where

$$\mu_1 = \min \left\{ 2k_V, 2k_\gamma, 2k_\alpha, 2k_P, \frac{p_{V,1}}{\lambda_{\max}(H_{V,1})}, \frac{p_{V,2}}{\lambda_{\max}(H_{V,2})}, \frac{p_{\gamma,1}}{\lambda_{\max}(H_{\gamma,1})}, \frac{p_{\gamma,2}}{\lambda_{\max}(H_{\gamma,2})}, \frac{p_{P,1}}{\lambda_{\max}(H_{P,1})}, \frac{p_{P,2}}{\lambda_{\max}(H_{P,2})}, \frac{p_{P,3}}{\lambda_{\max}(H_{P,3})}, \frac{p_{P,4}}{H_{P,4}} \right\}$$

$$\beta_1 = \sum_{i=1}^2 \frac{p_{V,i}}{2} \Theta_{V,i}^2 + \sum_{i=1}^2 \frac{p_{\gamma,i}}{2} \Theta_{\gamma,i}^2 + \sum_{i=1}^4 \frac{p_{P,i}}{2} \Theta_{P,i}^2 \quad (55)$$

From (53), we can get

$$L \leq \beta_1 / \mu_1 + [L(0) - \beta_1 / \mu_1] e^{-\mu_1 t} < +\infty \quad (56)$$

- (1) From (56) it is straightforward to show that L is bounded. If $z_\alpha \rightarrow -A_{z_\alpha}$, $z_\alpha \rightarrow B_{z_\alpha}$ or $|z_P| \rightarrow A_{z_P}$, then there is $L \rightarrow \infty$, which contradicts with the conclusion that L is bounded. Thus $-A_{z_\alpha} \leq z_\alpha \leq B_{z_\alpha}$, $|z_P| \leq A_{z_P}$ hold, namely $-A_\alpha \leq \alpha \leq B_\alpha$, $|P| \leq A_P$ hold.
- (2) From the conclusion that L is bounded and the inequalities $-A_{z_\alpha} \leq z_\alpha \leq B_{z_\alpha}$, $|z_P| \leq A_{z_P}$, it can be seen that all state tracking errors and parameter estimation errors ultimately converge into a bounded set.
- (3) Altitude command h_c , velocity command V_c , flight path angle command γ_c and the derivative of flight path angle command $\dot{\gamma}_c$ is known to be bounded, then the virtual command v_α and v_p , control input ER and δ_e , system states h, V, γ, α and P are all bounded, that is to say all the closed-loop signals are uniformly bounded.

4. Simulation and comparison results

In this section, numerical simulation is implemented to verify the effectiveness of the proposed method in Theorem 1. Considering the climb maneuver of AHV, the altitude command is to cruise at 24600 m for 50s, then climb to 27500 m, and the velocity command has been maintained at 1981.2 m/s. The aerodynamic parameters, propulsion system parameters of AHV in the simulation are shown in Table 1. The initial conditions and state constraints are shown in Table 2, the values of $-A_\alpha, B_\alpha$ and A_P are set referred to Refs. [17,19,20] and practical engineering application. The controller parameters are shown in Table 3. In order to more clearly show the effectiveness of the proposed method, an adaptive backstepping controller based on quadratic Lyapunov function is designed for comparison. The simulation results are shown in Figs. 3–19.

As can be seen from Figs. 3 and 4, which demonstrate the altitude and altitude tracking error, under the compared method, the altitude tracks altitude command well and the altitude tracking error is less than 3.1 m, while from 211.45s the angle of attack is out of the permitted range of the operation of scramjet engine. Under the proposed method, when the nominal virtual command of angle of attack $v_{\alpha,c}$ is within the limit of the asymmetric smooth saturation function, the altitude tracks altitude command well and the altitude tracking error is small. When the nominal virtual command of angle of attack $v_{\alpha,c}$ exceeds the limit of the asymmetric smooth saturation function ($kB_{v_\alpha} = 5.51$ deg) after 172.225s, the $v_{\alpha,c}$ generated by the altitude tracking error cannot be fully executed, so the altitude tracking error increases after 172.225s, but the angle of attack is within the allowable range of the operation of the scramjet engine all the time.

The velocity and velocity tracking error are given in Figs. 5 and 6, both compared method and proposed method track velocity command well, velocity tracking error tend to zero.

Flight path angle command, flight path angle and flight path angle tracking error are shown in Figs. 7–9. As can be seen from the figures, under the compared method, flight path angle command is very small all the time, and the flight path angle tracks flight path angle command well and the flight path angle tracking error is approximately zero. Under the

proposed method, the flight path angle command and flight path angle tracking error before 172.225s are approximately equal to the compared method, After 172.225s the flight path angle command begin to differ from the compared method, and then due to the further increase of the altitude tracking error, the flight path angle command also increases, because of the limit of the asymmetric smooth saturation function, the flight path angle cannot track the flight path angle command well, and the flight path angle tracking error also increases further.

Virtual angle of attack command, angle of attack and angle of attack tracking error are illustrated in Figs. 10–12. Under the compared method, it can be seen that the angle of attack exceeds B_α in 211.44s–350s, while under the proposed method, the angle of attack is within B_α all the time. This is because the virtual angle of attack command is limited to the range of $-A_{v_\alpha} \leq v_\alpha \leq B_{v_\alpha}$, the angle of attack tracking error is constrained to the range of $-A_{z_\alpha} \leq z_\alpha \leq B_{z_\alpha}$, so the angle of attack is within the permitted range of $-A_\alpha \leq \alpha \leq B_\alpha$.

Virtual angle of attack rate command, angle of attack rate and angle of attack rate tracking error are represented in Figs. 13–15. Under the compared method, it can be seen that the angle of attack rate exceeds A_p in 0.085s–0.18s, while under the proposed method, the angle of attack rate is within A_p all the time. This is because the virtual angle of attack rate command is limited to the range of $|v_p| \leq A_{v_p}$, the angle of attack rate tracking error is constrained to the range of $|z_p| \leq A_{z_p}$, so the angle of attack rate is within the permitted range of $|p| \leq A_p$.

Relevant control inputs under the two methods are shown in Figs. 16 and 17. As can be seen from the figures, the elevator and the equivalent ratio are within the allowed range.

Estimates of the adaptive law under the proposed controller are given in Figs. 18–20. As can be seen from the figures, $\hat{\Theta}_{V,2}^T \Phi_{V,2}$, $\hat{\Theta}_{\gamma,2}^T \Phi_{\gamma,2}$ and $\hat{\Theta}_{P,4}^T \Phi_{P,4}$ are non-zero, so ER , v_α and δ_e are nonsingular.

5. Conclusions

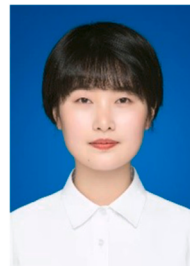
An adaptive backstepping control method with state constraints is proposed in this paper. The main novelty of this paper is to guarantee that the angle of attack and angle of attack rate are always within the operating limits of scramjet engine during hypersonic flight, and effectively avoid the occurrence of inlet unstart. Simulation results show that virtual commands are limited by smooth saturation function, tracking errors are constrained by BLF, so the desired state constraints are not violated, thus the effectiveness of the proposed method is proved. Future study will consider overcoming the requirement of the proposed method for the initial value.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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